

Treemaps and Choropleth Maps Applied to Regional Hierarchical Statistical Data

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Abstract

An emerging and challenging Geovisual Analytics application domain is visualization of hierarchical regional (sub-national) statistics. The OECD regional database is a potential treasure chest for policy-makers, researchers and citizens to gain a better understanding of a region's structure and performance and to carry out analysis of territorial trends and disparities based on sound information comparable across countries. New methods are needed to visually make comparisons between groups on different hierarchical levels, such as cities with countries or parts of countries. In this context, we demonstrate and reflect upon the potential synergy between information and geovisualization through the use of a squarified treemap dynamically linked to a choropleth map to facilitate visualization of complex hierarchical social science data. We exemplify and evaluate our approach with three usage scenarios that explore population change in the OECD countries using 1) squarified treemap, 2) regional choropleth map and 3) combined treemap and choropleth map.

Keywords--- Information visualization, treemap, choropleth map, linked views, demographics data

1. Introduction

Massive volumes of geospatial statistical data are today generated all over the world but not used as effectively as one would wish for [16]. Too little focus has been given to making advanced Geovisual Analytics technologies practical and accessible to statisticians. This paper reflects a challenging applied research task to stimulate, at global level, an exchange of best practices through Geovisual Analytics [18]. Tools are introduced that help establish progress initiatives at international and sub-national levels aimed at measuring economic, social and environmental developments. In this article we demonstrate and reflect upon the synergy between the treemap and the choropleth map. We illustrate the concept by visualizing hierarchical high resolution population data from OECD, with three usage scenarios:

- *Treemap only* - to explore a large hierarchical dataset containing population data in different age

groups and to discuss strengths and weaknesses of this method of representation.

- *Choropleth map only* – to discuss strengths and weaknesses of this method of representation and to make comparisons with the treemap.
- *Combined treemap and choropleth map* applied in a dynamically linked views environment – to illustrate how the two methods counter each other's main weaknesses when used together.

The main contribution of our paper is an informal evaluation with domain experts and the authors' informal reflections that a treemap representation could be a valid visual exploration and communication method for analyzing large hierarchical social science data. We also suggest that benefits can be gained by dynamically linking a choropleth map view with a treemap when searching for patterns and structure in the data. The paper is organized as follows: Related work is described below followed by an overview of the hierarchical regional OECD dataset in Section 3. Sections 4–6 describe the three application scenarios and their assessment. In the last section a longer analytical reasoning is given based on informal user evaluation and the authors' own conclusions.

2. Related work

Hierarchical data is often visualized as data dense treemap layouts [1], [2], which fill the screen efficiently without empty space between data items. Rectangles are created in a visual order that matches the input to a treemap algorithm [3], [4], where dense layouts recursively subdivide space at each level in a hierarchy and child nodes are placed inside their parents. Each data item is considered to be a node and can have labels and statistical attributes encoded as relative object size or colour. Treemaps provide spatial area for nodes under the same parent and visually emphasize the sizes of, in our application, “population” at all levels in the regional statistical data hierarchy. This is not difficult to program, can run fast and is easy to perceive and compare visually.

Shneiderman and Plaisant [6] give an excellent historical overview of the evolution and commercial success of treemaps. Geospatial hierarchical data applied in treemaps and grouped by space was first used by

Smartmoney [7] and later by Panopticon [8]. A related spatial treemap approach was recently introduced by Slingsby, Dykes and Wood [9], [10]. They consider variable combinations as a hierarchy and use treemaps for a large dataset and demonstrate links between treemap and maps. Little published research, however, has so far been made to use a combination of treemaps and maps to explore multilevel hierarchical geographical data.

Our approach demonstrates and reasons about the benefits of a single or combined choropleth map and treemap. We explore raw and aggregated social science regional data at several hierarchical levels aiming to discover statistical patterns that relate to significant characteristics of regions under study.

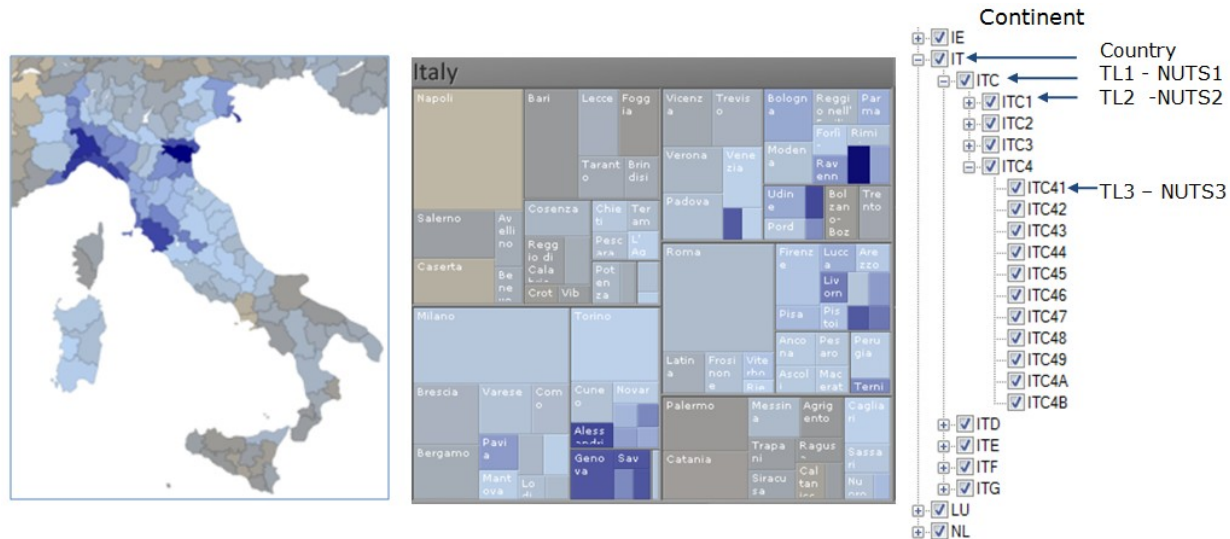


Figure 1: A squarified treemap ordered by population size and NUTS1 regions, dynamically linked to a choropleth map, both coloured by the ratio of the population that is older than 65 years, and applied on a limited (Italy) OECD regional hierarchical dataset. The spatial hierarchical data structure is based on 5 levels: continent, country, NUTS1, NUTS2 and NUTS3.

We are interested in finding answers about what methods and tasks are important when exploring demographical hierarchical data, such as general overview, trends over time, geographical patterns, indicator correlation, outliers and simultaneously mapping two dependent indicators such as age group and total population. For example, for a choropleth map screen space is always allocated depending on geographical area rather than an indicator of interest? Can the treemap compensate this weakness and are its known strengths and weaknesses applicable to the demographics data domain? A well-known approach for showing regional statistical data is the cartogram [14], which distorts a map by resizing its regions according to, for example, population density or total population. We evaluated this method for OECD regional data but the domain experts could not recognize the many small regions and found it almost impossible to tell which of two regions of similar size but different shape is the largest, or by how much the size differs.

Many papers [11], [12], [13] describe a combination of choropleth map, scatter plot, parallel coordinates plot and heatmap etc. These papers do, however, not address the need for access to regional datasets on all hierarchical levels, to for example make comparisons between groups on different hierarchical levels, such as cities with countries or parts of countries.

3. Dataset

OECD countries have experienced a growing interest in regional development [15]. The performance of regional economies and the effectiveness of regional policy help determine a nation's growth and shape the measure of well-being across countries. The *OECD Regional database* [16] provides statistical information at sub-national level for all the OECD member countries. It contains yearly time-series for around 50 indicators on demography, economy and labour market opportunities, environment, social issues and innovation related activities for more than 1,700 regions of the OECD countries. Data is organized in a five-level hierarchical structure – continent, country, Territorial Level 1 (TL1), TL2 and TL3 or European corresponding NUTS(1,2,3) regions. These levels are illustrated in Figure 1. For this usage scenario, each region contains information about population 2002-2007, for each year and separated by the age groups 0-14, 15-64 and 65+. The population values also let us calculate the average population over time, average yearly population change in percent (the average yearly percentage increase or decrease is calculated using regression analysis to fit an exponential curve to the population measurements for each year) and the percentage of the population that falls into each age group.

4. Treemap Displaying Population Change

Our first usage scenario, figure 3, presents a treemap built on the full OECD regional dataset, using the hierarchy Age Group, Continent, Country, TL1, TL2, Region Name. The size of each cell represents the average population of each region during the 2002-2007 periods and the colour represents the population change per year in percent. Below the treemap is a colour legend with an overlaid histogram showing the population change distribution. To highlight highs and lows without making the representation overly sensitive to outliers, the colours are scaled to match this distribution using the 1st, 10th, 90th and 99th percentiles and with gray centred at zero (no change). Red tones represent a growing population and blue tones represent a shrinking population.

The top half of the treemap displays values for the total population, while the bottom half is split into the age groups 0-14, 15-64 and 65+. This means that Japan for instance shows up four times; once with its total population displayed in the top half and three times in the bottom half, where it is split into the three age groups. Japan's total cell area is the same in the bottom and top half. Note that a region's rate of change for a specific age group corresponds to the size of that age group only, not to its fraction of the region's total population.

The software used to create figure 2 lets the user interact by selecting different regions in the treemap to compare their exact values in a tabular view. The user can also drill down in the data by choosing to render only a specific branch of the tree, for instance ages 0-14, Europe, Germany. This is done by double clicking on a group in the plot, which then allocates the entire screen area to that branch so that smaller items can be identified.

4.1. What can be seen?

We see from the size of the European group (top left quarter of the treemap in figure 2) that the population of the European OECD member countries together is slightly larger than that of North America. We also see that while there are slightly more children (people aged 0-14) in North America than in Europe and far less in Japan, Europe has more elderly people (ages 65+) than North America and Japan together. Japan also has around 70% as many elderly people as the USA, despite that its total population is less than half the size.

Turning our attention to the colour of the cells in the top half of the treemap (the total population), we see far more red tones than blue. This indicates that the OECD population as a whole is generally increasing, which is correct. Such a conclusion can be drawn in a treemap where colour is mapped to the rate of change in size of each cell, but would be impossible to draw in a choropleth map where the size instead represents the physical area of each region.

We also see that the number of children is decreasing overall in the OECD member countries and that the number of elderly people is increasing. Turkey, Poland, Germany and Korea are seeing dramatic decreases in child populations with numbers in parts of Turkey decreasing by up to eight percent per year. Note however that there are very large differences between different regions in Turkey, which we will return to in the next section. Just like the young population is decreasing, Turkey, Korea, Mexico, Japan, Germany and many other European countries face equally dramatic increases in elderly populations, again with Turkey taking the lead with yearly increases of up to 14%.

A few regions show values that drastically differ from all other regions within the same country. This might indicate changes in regional borders, errors in the data or important local events. The latter is likely the case for New Orleans where the hurricane Katrina forced a large part of the population to move out in August 2005. If we look closely, we can even see that New Orleans' population decrease is much larger for children than for senior citizens.

Last but not least, the treemap representation makes it very obvious just how small many of the European countries are population wise. In fact, half of these countries have smaller populations than the largest American, Japanese, Turkish and Korean city regions. The use of map borders for the treemap structure does however also mean that political decisions on how to draw those borders affects our interpretation of the data. An example of this is that Paris is shown as having roughly the same population as Stockholm, while in reality the urban populations of the two cities have a ratio of around 8:1.

5. Subset of Data in Choropleth Map

For reference we have included a subset of the data from figure 2 visualized using standard choropleth maps. Figure 3 shows the yearly population change for the regions in the European OECD member countries, for each age group.

It is safe to say that the geographical representation is more intuitive for most people to interpret than the treemap is. The geographical patterns that were lost in the treemap also show up, with Turkey having very large increases in the number of senior citizens mainly in the south-eastern (Kurdish) regions. The maps also show that the number of children in Italy is decreasing in the south and increasing in the north.

However, this perceived clarity does come at a cost. The maps only visualize one indicator of interest (population change) and all information about absolute population size is lost. This for instance makes it easy to draw the false conclusion that the total population of Turkey is decreasing, as there is more blue than red on the map. In reality, Turkey experiences a slight yearly population increase. The treemap visualization in figure 4 does not suffer from the same problem and we can see

that Istanbul (a very small region on the map) is inhabited by a very large (and growing) part of the total Turkish population.

6. Treemap and Choropleth Map Together

Our last usage scenario, figure 4, shows the young population of the European OECD member countries visualized using a combination of a choropleth map and a treemap. Colour in this figure, which again is mapped using the 1st, 10th, 50th, 90th and 99th percentiles, represents the percentage of the total population that falls into the 0-14 age group. In other words, red areas have a high ratio of children and blue areas have a low ratio. Cell size in the treemap represents the total number of people in each region.

Four extreme clusters of regions immediately stand out in the choropleth map. Again, south-eastern Turkey is one of them, with children in some parts making up almost half the population¹. On the opposite end of the distribution, with ratios down to nine percent, lie former East Germany, north-eastern Spain and northern Italy.

The additional information provided by the treemap that the choropleth map lacks is all based on the introduction of a second indicator; in this case the size of each region's population. For instance, it becomes apparent that despite that Sweden is the third largest of the included countries measured in physical area, it only has a population of nine million – less than Turkey's Istanbul region alone. It is also possible to see that while Germany and Turkey currently have almost identical total population numbers, Turkey has far more children. Over the next few decades Turkey is therefore likely to overtake Germany as the European OECD member country with the highest population.

7. Conclusions

We have with three usage scenarios shown that the treemap is a valid visualization method for hierarchical demographics data and that many benefits are gained when linking a choropleth map view with a treemap. Domain experts who have analyzed this data use the choropleth map's geographical representation to draw conclusions that take into account political and historical events that may have caused the current trends.

At the same time, the treemap offers an alternative way of simultaneously viewing the same data, independent of geography. In this view it is much easier to compare groups of regions, even on different hierarchical levels. This is true in particular when mapping cell size to an absolute number and colour to the rate of change of the same indicator, as an average rate of change can be estimated based on the average colour of the group. Doing this in a choropleth map can lead to highly incorrect conclusions, as a tiny region may

(and in the case of densely populated capital regions often does) account for almost the entire change.

Our first informal usability tests with domain experts resulted in both positive and negative results but encouraging qualitative feedback. We summarize below some comments based on informal discussion and evaluation between domain experts and authors:

- It is easier to generate hypotheses from a choropleth map than from a treemap, but hypotheses generated from a treemap may hold more value.
- The treemap is difficult to interpret, but even more difficult to misinterpret. Once a pattern has been found it is very rarely later dismissed as incorrect.
- The choropleth map is easy both to interpret and to misinterpret, as many conclusions fail to take into account that indicator values for different regions may not correlate with region size.
- Colour and area that dominate the treemap are fairly vague value mappings. This means that the treemap can give an overview and be hypothesis generating, but that other methods which give the user access to more precise comparisons are required for validation.
- The treemap is a rather complex visual representation that requires the user to focus intensely on it to be able to fully draw conclusions. This should be taken into consideration when combining it with other views, as there is a risk for information overload. Our experiences show that users have a tendency to look at the views that are easier to understand and draw conclusions from there, instead of trying to interpret the treemap.
- Although the treemap-choropleth map combination may lack the strong visual impact that often characterizes a cartogram, we argue that it gives a truer image of the visualized data and that it is therefore far more suitable for demographical analytical situations.
- All hypothesis evaluation related to social science data requires a strong understanding of the underlying factors that govern the measured indicators, but the evaluation process can efficiently be supported by a combination of these examples and other methods that let the analyst access the values directly.
- Statistical social science data collected per region benefits strongly from the alternative viewpoint provided by the treemap representation. Regions of high interest tend to be characterized by high population density; or a large population (or otherwise large absolute indicator values) and a small geographical size. Such characteristics are allocated more screen space in a treemap, unlike a choropleth map where the relation is inverted and population dense regions tend to be very small.

We have reached the conclusion based on our own reasoning together with domain experts' evaluation that

¹ Less than four percent of the population is above 65 years of age in many of the same regions.

for a combined treemap and choropleth map approach the analyst is able to:

- Easily see geographical patterns and clusters, including ones that span national borders.
- Make far more accurate estimates of the average values for groups of regions, than if a choropleth map had been used alone.
- In a single view compare individual regions and groups of regions with each other on different hierarchical levels.
- Focus the attention to regions of interest, even when they are geographically very small.

Our next step is to conduct a more controlled evaluation within our engagement in the Global Project on "Measuring the Progress of Societies" [18], which seeks to become the world wide reference point for those who wish to assess the progress of their societies.

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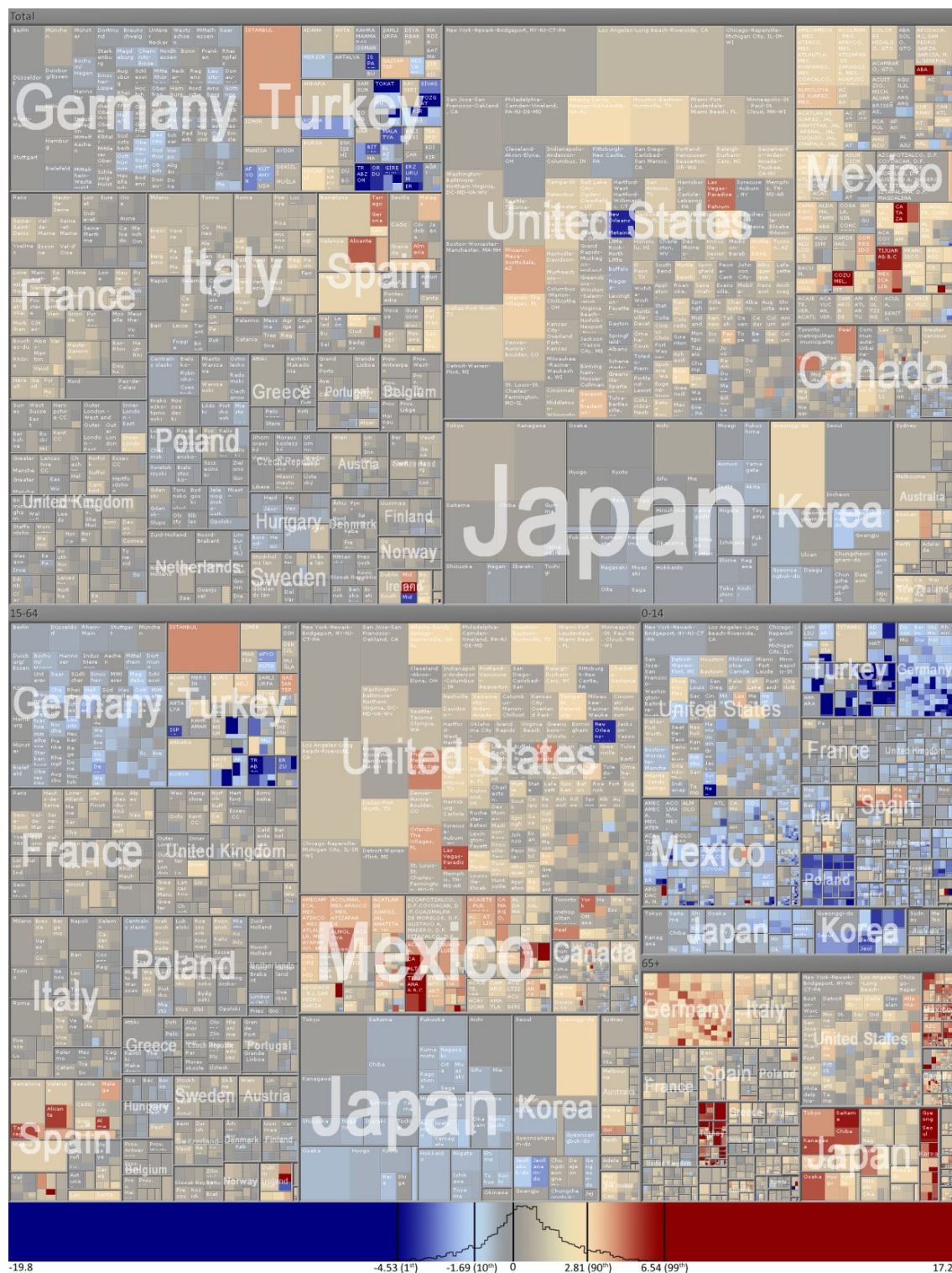


Figure 2: The population and population change in the OECD member countries, plotted as a Treemap. Colour represents population increase (red) or decrease (blue) and the size of each cell is mapped to the average population of each region during the measured time period. The top half of the Treemap shows the values for the total population, while the bottom half is split into three different age groups 0-14, 15-64 and 65+. The image is available in colour and full resolution at: http://ncvta.itn.liu.se/content/1/c6/14/52/52/Pictures/oecd_population_change_treemap_A4%20FIGURE%203.png

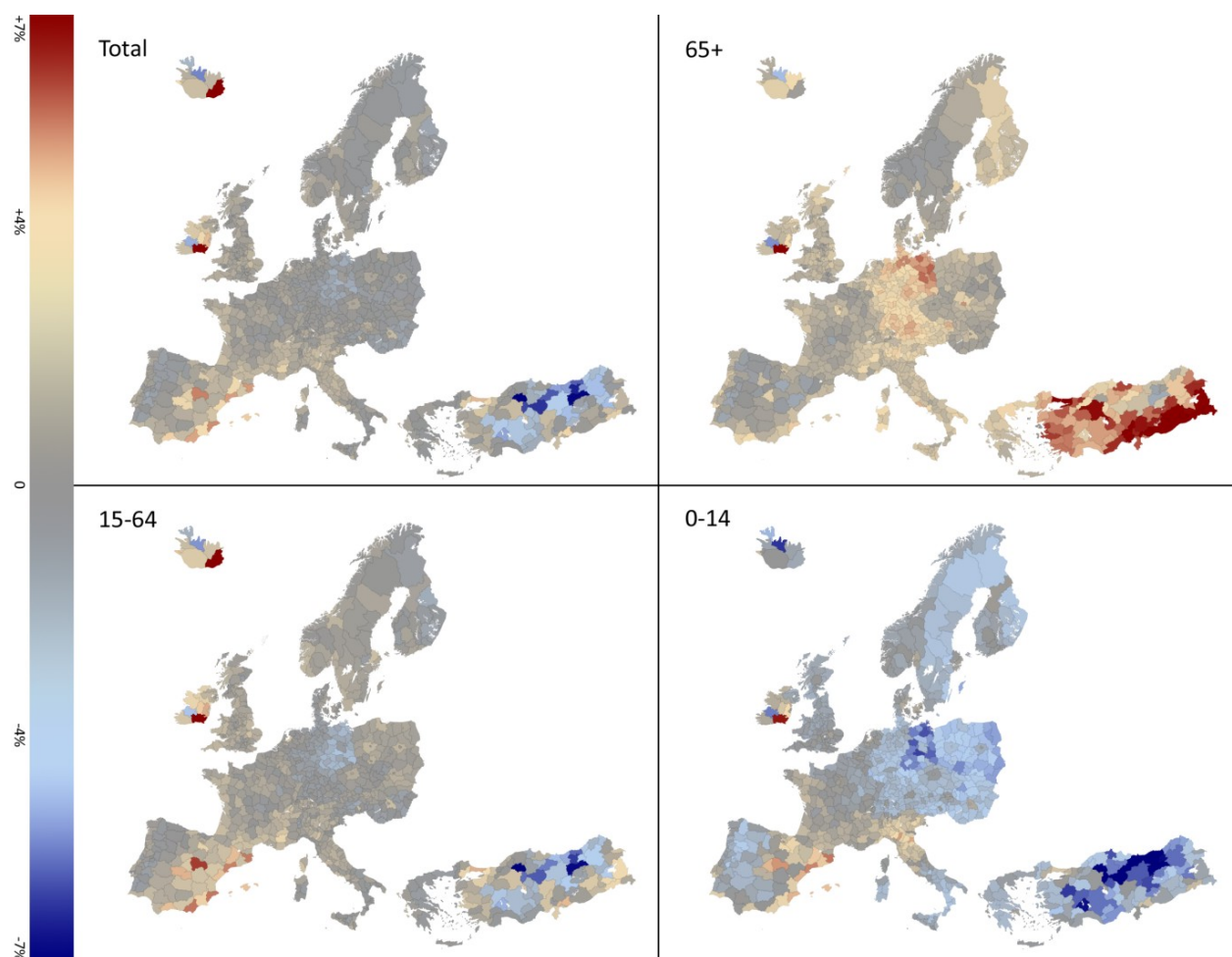


Figure 3: The estimated yearly population change in percent in the regions of the European OECD member countries. Each map shows the change within a specific age group.

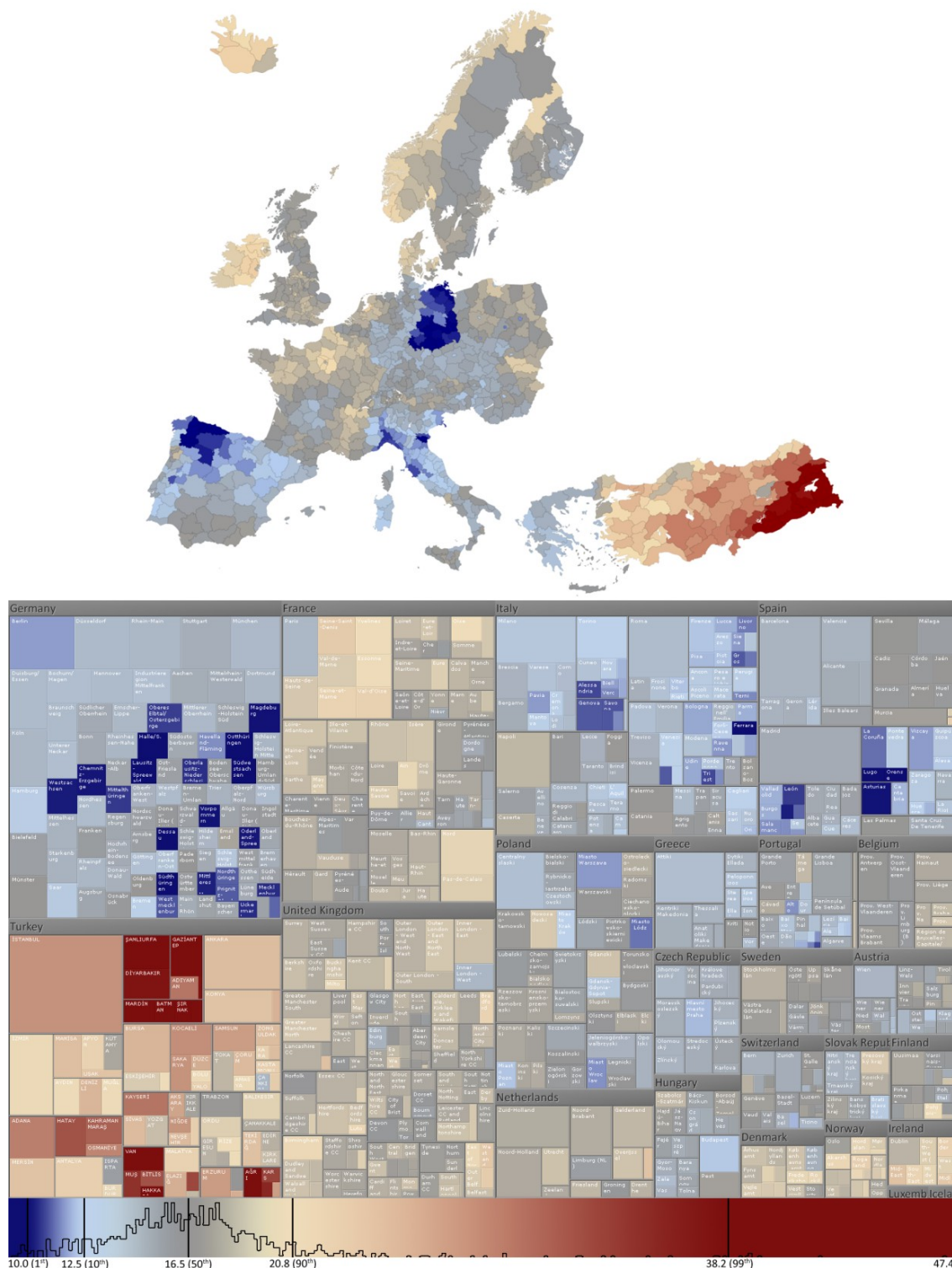


Figure 4: Linked treemap and choropleth map showing the ratio of children in the European OECD member countries. The colour of each region represents the percentage of the total population that falls within the 0-14 age group. Size in the Treemap shows the size of the total population. The image is available in colour and full resolution at: http://ncv.a.itn.liu.se/content/1/c6/14/52/52/Pictures/europe_0-14_ratio_treemap%2Bchoropleth_A4%20Figure%205.png.